

CODE OF DATA visualization

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import matplotlib.patches as mpatches
from matplotlib.lines import Line2D

df = pd.read_csv(r"C:\Users\nayan\OneDrive\Desktop\All_data.csv") # here you can change
the path of your csv file

print(df.head())
print()
print("print mathematical calculation")
print(df.describe())
print()
print("no. of column and rows and column name, Data type and non null and null value
display\n")
print(df.info())
print("show no. of rows ans columns")
print("Shape:", df.shape)

plt.figure(figsize=(12, 6))
plt.ylim(0, 100)
plt.yticks(np.arange(0, 110, 10))

df['Date'] = pd.to_datetime(df['Date'])

plt.xlabel('Date')
plt.ylabel('PR%')
plt.title('Performance Ratio Evolution\n From 2019-07-01 to 2022-03-24')
plt.grid(True)

def get_color(ghi):
    if ghi < 2:
        return 'navy'
    elif 2 <= ghi < 4:
        return 'lightblue'
    elif 4 <= ghi < 6:
        return 'orange'
    else:
        return 'brown'
```

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plt.scatter([], [], marker='D', color='navy', label='< 2', s=5)
plt.scatter([], [], marker='D', color='lightblue', label='2 - 4', s=5)
plt.scatter([], [], marker='D', color='orange', label='4 - 6', s=5)
plt.scatter([], [], marker='D', color='brown', label='> 6', s=5)

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plt.legend(title='Daily Irradiation [kWh/m²]',loc='upper left',ncol=5,frameon=False)

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target_pr = 73.9

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points_above = (df['PR'] > target_pr).sum()
total_points = len(df)

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percentage = (points_above / total_points) * 100

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avg_7 = df['PR'].tail(7).mean()
avg_30 = df['PR'].tail(30).mean()
avg_60 = df['PR'].tail(60).mean()
avg_90 = df['PR'].tail(90).mean()
avg_365 = df['PR'].tail(365).mean()

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plt.text(
    0.3, 0.3,
    f'Points above Target PR = {points_above}/{total_points} = {percentage:.1f}%',
    transform=plt.gca().transAxes,
    fontsize=10
)

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summary_text = f"""
Avg PR last 7d : {avg_7:.1f} %
Avg PR last 30d : {avg_30:.1f} %
Avg PR last 60d : {avg_60:.1f} %
Avg PR last 90d : {avg_90:.1f} %
Avg PR last 365d: {avg_365:.1f} %
"""

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plt.text(0.82,0.02,summary_text,transform=plt.gca().transAxes, fontsize=9)

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df['Color'] = df['GHI'].apply(get_color)

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plt.scatter(df['Date'],df['PR'],marker='D',c=df['Color'],s=5,alpha=0.7)

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start_budget = 73.9
annual_drop = 1.0
start_year = 2019

def get_budget_pr(date):

    if date.month >= 7:
        fy_start = pd.Timestamp(f'{date.year}-07-01')
        year_index = date.year - start_year
    else:
        fy_start = pd.Timestamp(f'{date.year-1}-07-01')
        year_index = date.year - start_year - 1

    fy_start_budget = start_budget - (year_index * annual_drop)

    days_passed = (date - fy_start).days

    frac = days_passed / 365.25

    return fy_start_budget - (frac * annual_drop)

df['Budget_PR'] = df['Date'].apply(get_budget_pr)

plt.plot(df['Date'],df['Budget_PR'],color='darkgreen', linewidth=2,label='Target Budget PR')

df['rolling_mean'] = df['PR'].rolling(30, min_periods=1).mean()

plt.plot(df["Date"], df["rolling_mean"], color="red",linewidth=2,label="30 Day Average")

plt.xlim(pd.Timestamp("2019-07-01"), df["Date"].max())

plt.gca().xaxis.set_major_locator(mdates.MonthLocator(interval=3))
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter("%b/%y"))

plt.legend()
plt.tight_layout()
plt.show()

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if date.month >= 7:
    year_index = date.year - start_year
else:
    year_index = date.year - start_year - 1

return start_budget - (year_index * reduction_per_year)

df['Budget_PR'] = df['Date'].apply(get_budget_pr)
df['rolling_mean'] = df['PR'].rolling(30,min_periods=1).mean()

plt.plot(df['Date'],df['Budget_PR'],color='darkgreen',linewidth=2,
         label='Target Budget PR')
plt.plot(df["Date"], df["rolling_mean"], color="red",linewidth=2)
plt.xlim(pd.Timestamp("2019-07-01"),df["Date"].max())
plt.gca().xaxis.set_major_locator(mdates.MonthLocator(interval=3))
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%b/%y'))

plt.show()
```